Non-Commutativity in Geodesic Congruences

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(on work in collaboration with Prof. Louis Kauffman)

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Outline

Motivation

Geodesic Congruences, Raychaudhuri Equation and Singularity Theorems

Non-commutativity in Geodesic Congruences

Modified Focusing and Sigularity Theorems, the NCEC

Applications to Cosmology

Conclusions

A short history: from Einstein to Raychaudhuri

- **1907-15:** Einstein's field equations tie spacetime curvature to the matter-energy content.
- **1916:** Schwarzschild's solution already contained what appeared to be singular behavior. The singularity at $r = 2GM/c^2$ was regarded as a coordinate artifact, while the physical significance of the $r = 0$ singularity was initially subject to skepticism.
- **1922:** Friedmann: homogeneous, isotropic cosmological solutions. A finite-age, expanding universe becomes mathematically possible.
- **1955:** Raychaudhuri derives a single equation for the evolution of the expansion θ of any geodesic congruence. This is the technical engine behind every singularity theorem that follows.

Raychaudhuri's equation transformed the question whether one prove that, under sufficiently general and physically reasonable conditions, are singularities an inevitable feature of spacetime?

Penrose, Hawking, and the singularity theorems

- **1965:** Penrose: gravitational collapse of a trapped surface forces geodesic incompleteness
- **1966:** Hawking: develops singularity theorems showing that, under suitable energy and global conditions, an expanding timelike congruence is necessarily geodesically incomplete.
- **1967:** Hawking: further develops the causal-structure ingredients of the singularity theorems and shows how singularity results can be obtained without assuming a global Cauchy surface in appropriate circumstances.
- **1970:** Hawking & Penrose: the general singularity theorem.

Spacetime non-commutativity: the usual picture

- Quantum gravity approaches suggest the geometry of spacetime may need modification at high energies.
- Non-commutativity has been motivated by
  - Connes' non-commutative geometry.
  - deformation quantization, matrix models,
  - string theory, non-commutative QFT,
- In Connes' approach, the algebra of smooth functions on the manifold is replaced by a non-commutative algebra.
- The emphasis is on non-commuting coordinates : the smooth manifold is no longer fundamental, and contact with classical gravitational physics is less direct.

A short history of spacetime non-commutativity: early ideas

- **1930s:** Heisenberg proposed to Rudolf Peierls that non-commuting position coordinates could introduce a fundamental length scale and eliminate short-distance singularities. The idea subsequently circulated through Peierls, Pauli, and Oppenheimer, eventually reaching Oppenheimer's student Hartland Snyder.
- **1947:** Snyder first published a non-commutative spacetime model. Quantized coordinates, $[\hat{x}^\mu, \hat{x}^\nu] \neq 0$, while preserving Lorentz invariance.
- Motivation at the time: get rid of short distance divergences geometrically.

Deformation quantization and Connes' geometry

- **1946, 1949:** Groenewold and Moyal developed a formulation of quantum mechanics in which ordinary multiplication of phase space functions is replaced by the non-commutative Groenewold Moyal star product.
- **1991, 1994:** Connes developed non-commutative geometry into a systematic geometric framework: instead of describing a space through the commutative algebra $C^\infty(M)$ of functions on a manifold, one allows the underlying algebra $\mathcal{A}$ to be non-commutative.

String theory and noncommutative field theory

- **1999:** Seiberg & Witten showed that open strings ending on D-branes in a background B-field lead, in an appropriate low-energy limit, to gauge theory on a non-commutative space, establishing a direct connection between string theory and non-commutative field theory.
- **Common theme:** In these approaches, the classical notion of commuting spacetime coordinates can be replaced by an algebraic structure in which the coordinate degrees of freedom need not commute, $[x^\mu, x^\nu] = i\theta^{\mu\nu}$.

Where this work departs

- Every approach above relocates non-commutativity into the algebra of coordinates or functions on spacetime, the smooth manifold itself is replaced.
- This work keeps M a smooth manifold, with nice properties throughout.
- Instead, the non-commutativity is relocated to the kinematics of a geodesic congruence: $[U, V]^\mu \neq 0$, justified via the Frobenius theorem rather than an algebraic postulate.
- Payoff: the entire classical machinery, Raychaudhuri's equation, energy conditions, singularity theorems, stays available to be modified, rather than discarded.

Geodesic congruences

- Smooth congruence of timelike geodesics with tangent U^μ, $U^\mu U_\mu = -1$, and

$$B_{\mu\nu} = \nabla_\mu U_\nu, \qquad B_{\mu\nu}U^\mu = B_{\mu\nu}U^\nu = 0.$$

- Deviation vector V^μ between neighboring geodesics. Conventionally the tangent and deviation vectors commute,

$$\mathcal{L}_U V^\mu = [U, V]^\mu = 0 \;\Rightarrow\; U^\mu \nabla_\mu V^\nu = B^\nu{}_\mu V^\mu.$$

- Irreducible decomposition with $P_{\mu\nu} = g_{\mu\nu} + U_\mu U_\nu$:

$$B_{\mu\nu} = \tfrac{1}{3}\theta P_{\mu\nu} + \sigma_{\mu\nu} + \omega_{\mu\nu},$$

expansion θ, shear $\sigma_{\mu\nu}$, twist $\omega_{\mu\nu}$.

The Raychaudhuri equation

Timelike congruence (τ = proper time):

$$\frac{\mathrm{d}\theta}{\mathrm{d}\tau} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}U^\mu U^\nu.$$

Null congruence (λ = affine parameter, 2d transverse space):

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\frac{1}{2}\theta^2 - \hat{\sigma}_{\mu\nu}\hat{\sigma}^{\mu\nu} + \hat{\omega}_{\mu\nu}\hat{\omega}^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu.$$

- Shear term is non-positive; twist vanishes for hypersurface-orthogonal congruences.

Energy conditions and focusing

With Einstein's equations, $R_{\mu\nu}U^\mu U^\nu = 8\pi G\big(T_{\mu\nu}U^\mu U^\nu + \tfrac{1}{2}T\big)$.

- Strong energy condition $\Rightarrow R_{\mu\nu}U^\mu U^\nu \geq 0$; null case $R_{\mu\nu}k^\mu k^\nu \geq 0$.
- Hypersurface orthogonality then gives $\dfrac{\mathrm{d}\theta}{\mathrm{d}\tau} + \dfrac{\theta^2}{3} \leq 0$, i.e. $\dfrac{\mathrm{d}}{\mathrm{d}\tau}(\theta^{-1}) \geq \dfrac{1}{3}$.

Lemma (focusing). If $\theta_0 < 0$ at some point, then $\theta \to -\infty$ within

$$\tau \leq \frac{3}{|\theta_0|} \quad \text{(timelike)}, \qquad \lambda \leq \frac{2}{|\theta_0|} \quad \text{(null)}.$$

Conjugate points

- Writing $V^\mu(\tau) = A^\mu{}_\nu(\tau)\dot{V}^\nu(0)$ gives $\dot{A} = BA$ and

$$\theta = \mathrm{Tr}\big(A\dot{A}^{-1}\big) = \frac{d}{d\tau}\ln|\det A|.$$

- q is conjugate to $p \iff \theta \to -\infty$ at q.

Proposition 2.3

Let $(M, g_{\mu\nu})$ be a spacetime satisfying $R_{\mu\nu}U^\mu U^\nu \geq 0$ for all timelike U^μ. Let γ be a timelike geodesic and let $p \in \gamma$. Suppose the convergence of the congruence of timelike geodesics emanating into the future from p attains the negative value θ_0 at $r \in \gamma$. Then within proper time $\tau \leq 3/|\theta_0|$ from r along γ there exists a point q conjugate to p, assuming, of course, that γ extends that far.

Extremal length

- For a smooth one-parameter family of timelike curves $\lambda_\alpha(t)$ from p to q, with tangent T^μ and deviation X^μ orthogonal to T^μ, the second variation of proper time is

$$\left.\frac{d^2\tau}{d\alpha^2}\right|_{\alpha=0} = \int_a^b X^\nu \{T^\rho \nabla_\rho (T^\mu \nabla_\mu X_\nu) + R^\sigma{}_{\rho\mu\nu} T^\mu T_\sigma X^\rho\}\, dt = \int_a^b X^\nu (\mathcal{O}X)_\nu\, dt.$$

Theorem 2.4

Let γ be a smooth timelike curve connecting two points $p, q \in M$. Then the necessary and sufficient condition that γ locally maximizes the proper time between P and q over smooth one-parameter variations is that γ be a geodesic with no point conjugate to p between p and q.

Theorem 2.6

Let γ be a smooth timelike curve connecting a point $p \in M$ to a point q on a smooth spacelike hypersurface Σ. Then the necessary and sufficient condition that γ locally maximizes the proper time between p and Σ over smooth one-parameter variations is that γ be a geodesic orthogonal to Σ with no conjugate point to Σ between Σ and p.

Curvature of a Cauchy surface

- For U^μ the unit tangent field of the congruence of timelike geodesics orthogonal to a spacelike hypersurface Σ, the extrinsic curvature is

$$K_{\mu\nu} = \nabla_\mu U_\nu = B_{\nu\mu}, \qquad K_{\mu\nu}U^\mu = K_{\mu\nu}U^\nu = 0, \qquad K_{\mu\nu} = \tfrac{1}{2}\mathcal{L}_U h_{\mu\nu}.$$

- In Gaussian normal coordinates $K_{\mu\nu} = \frac{1}{2}\partial h_{\mu\nu}/\partial t$: the "time derivative" of the induced spatial metric.
- The trace $K = K^\mu{}_\mu = h^{\mu\nu}K_{\mu\nu}$ satisfies $K = \theta$.
- p is conjugate to Σ along γ iff the expansion θ of the congruence orthogonal to Σ approaches $-\infty$ at p.

Proposition 2.5

Let $(M, g_{\mu\nu})$ be a spacetime satisfying $R_{\mu\nu}U^\mu U^\nu \geq 0$ for all timelike U^μ. Let Σ be a spacelike hypersurface with $K = \theta < 0$ at a point $q \in \Sigma$. Then within proper time $\tau \leq 3/|K|$ there exists a point p conjugate to Σ along the geodesic γ orthogonal to Σ and passing through q, assuming that γ can be extended that far.

Singularity Theorem I: Timelike Incompleteness

Theorem 2.15

Let $(M, g_{\mu\nu})$ be a globally hyperbolic spacetime with $R_{\mu\nu}U^\mu U^\nu \geq 0$ for all timelike U^μ, which will be the case if Einstein's equation is satisfied with the strong energy condition holding for matter. Suppose there exists a smooth (or, at least C^2) spacelike Cauchy surface Σ for which the trace of the extrinsic curvature (for the past-directed normal geodesic congruence) satisfies $K \leq C < 0$ everywhere, where C is a constant. Then no past-directed timelike curve from Σ can have length greater than $3/|C|$. In particular, all past-directed timelike geodesics are incomplete.

Proof idea. Suppose a past-directed timelike curve λ from Σ had length $> 3/|C|$. A maximum-length curve γ to a point p beyond that length would, by Theorem 2.12, be a geodesic with no conjugate point between Σ and p, contradicting Proposition 2.5. Hence λ cannot exist.

Singularity Theorem II: Null Incompleteness

A compact, two-dimensional, smooth spacelike submanifold T, having the property that the expansion θ of both sets (i.e., "ingoing" and "outgoing") of future-directed null geodesics orthogonal to T is everywhere negative, is called a *trapped surface*.

Theorem 2.16

Let $(M, g_{\mu\nu})$ be a connected, globally hyperbolic spacetime with a noncompact Cauchy surface Σ. Suppose $R_{\mu\nu}k^\mu k^\nu \geq 0$ for all null k^μ, as will be the case if $(M, g_{\mu\nu})$ is a solution of Einstein's equation with matter satisfying the strong energy condition. Suppose further that M contains a trapped surface T. Let $\theta_0 < 0$ denote the maximum value of θ for both sets of orthogonal geodesics in T. Then at least one inextendible future-directed orthogonal null geodesic from T has affine length no greater than $2/|\theta_0|$.

Relaxing $[U, V] = 0$: the Frobenius theorem

- A distribution D is involutive if $[X, Y] \in D$ whenever $X, Y \in D$.
- **Frobenius:** For a smooth, C^∞ manifold $\mathcal{M}$, D is an involutive, C^∞ distribution of $\mathcal{M} \iff$ the distrubtution $\mathcal{D}$ is integrable.
- Here $D = \mathrm{span}(U, V)$. Even when it is integrable, the commutator need not vanish:

$$\mathcal{L}_U V^\mu = [U, V]^\mu = \kappa^{\mathrm{NC}} U^\mu + \frac{\theta^{\mathrm{NC}}}{3} V^\mu.$$

- κ^{NC}: component along the tangent; θ^{NC}: component along the separation.
- Deviation vectors differing by a multiple of U^μ describe the same neighbouring geodesic, so we drop the longitudinal piece:

$$[U, V]^\mu = \frac{\theta^{\mathrm{NC}}}{3} V^\mu.$$

Effective deformation tensor

The evolution of the separation vector becomes

$$\frac{DV^\mu}{\mathrm{d}\tau} = U^\nu \nabla_\nu V^\mu = \big(B^\mu{}_\nu + \frac{\theta^{\mathrm{NC}}}{3} P^\mu{}_\nu\big) V^\nu.$$

- $B^\mu{}_\nu$ alone no longer characterizes the congruence. Define

$$\tilde{B}^\mu{}_\nu = B^\mu{}_\nu + \frac{\theta^{\mathrm{NC}}}{3} P^\mu{}_\nu = \tfrac{1}{3}(\theta + \theta^{\mathrm{NC}}) P^\mu{}_\nu + \sigma^\mu{}_\nu + \omega^\mu{}_\nu.$$

- Shear and twist are unchanged; only the expansion is shifted:

$$\tilde{\theta} = \theta + \theta^{\mathrm{NC}}$$

- θ^{NC} acts as a free geometric function characterizing the congruence.

The modified Raychaudhuri equation

$$\frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}\tau} = -\frac{(\tilde{\theta} - \theta^{\mathrm{NC}})^2}{3} - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}U^\mu U^\nu + \frac{\mathrm{d}\theta^{\mathrm{NC}}}{\mathrm{d}\tau}$$

- Same structure as the standard equation, but the physically relevant expansion is $\tilde{\theta}$, not θ.
- Equations for $\sigma_{\mu\nu}$ and $\omega_{\mu\nu}$ retain their standard form and interpretation.
- Two extra source terms: θ^{NC} dependent terms from the shifted square, and $\dot{\theta}^{\mathrm{NC}}$.

Expanding the square,

$$\frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}\tau} + \frac{\tilde{\theta}^2}{3} = -R_{\mu\nu}U^\mu U^\nu + \underbrace{\frac{2}{3}\theta^{\mathrm{NC}}\tilde{\theta} - \frac{(\theta^{\mathrm{NC}})^2}{3} + \dot{\theta}^{\mathrm{NC}}}_{\mathcal{E}^{\mathrm{NC}}} - \sigma^2 + \omega^2.$$

Focusing under the strong energy condition

- Naively dividing by $\tilde{\theta}$ gives

$$\tilde{\theta}^{-1} \geq \tilde{\theta}_0^{-1} + \frac{\tau}{3} - \int \Big(\frac{2\theta^{\mathrm{NC}}}{3\tilde{\theta}} + \frac{1}{\tilde{\theta}^2} \frac{\mathrm{d}\theta^{\mathrm{NC}}}{\mathrm{d}\tau} \Big) \mathrm{d}\tau,$$

  and for generic θ^{NC} the integral has no definite sign.
- But the commutative expansion $\theta = \tilde{\theta} - \theta^{\mathrm{NC}}$ still obeys $\dot{\theta} + \theta^2/3 \leq 0$ under the SEC, so it focuses within $\tau \leq 3/|\theta_0|$.
- Hence, if θ^{NC} stays finite (bounded above) along the geodesic up to the focusing point,

$$\tilde{\theta} = \theta + \theta^{\mathrm{NC}} \to -\infty \quad \text{within the same proper time.}$$

- The SEC is sufficient for focusing of $\tilde{\theta}$ in this case, but no longer necessary.

The Non-Commutative Energy Condition (NCEC)

NCEC

$$R_{\mu\nu}U^\mu U^\nu \;\geq\; \mathcal{E}^{\mathrm{NC}}, \qquad \mathcal{E}^{\mathrm{NC}} = \frac{2\theta^{\mathrm{NC}}\tilde{\theta}}{3} + \frac{\mathrm{d}\theta^{\mathrm{NC}}}{\mathrm{d}\tau} - \frac{(\theta^{\mathrm{NC}})^2}{3}.$$

Substituting into the modified Raychaudhuri equation gives

$$\frac{\mathrm{d}\tilde{\theta}}{\mathrm{d}\tau} + \frac{\tilde{\theta}^2}{3} \leq 0.$$

- $\mathcal{E}^{\mathrm{NC}} > 0$: stronger than the SEC. $\mathcal{E}^{\mathrm{NC}} = 0$: coincides with the SEC.
- $\mathcal{E}^{\mathrm{NC}} < 0$: **weaker** than the SEC, so it can hold even when $R_{\mu\nu}U^\mu U^\nu < 0$.
- Focusing of $\tilde{\theta}$, and hence geodesic incompleteness, can persist in regimes where the standard Penrose Hawking analysis doesn't hold up.

Generalized focusing lemmas

Lemma (timelike). $\tilde{\theta} \to -\infty$ in finite proper time if either

- $R_{\mu\nu}U^\mu U^\nu \geq 0$, $\tilde{\theta}_0 < \theta_0^{\mathrm{NC}}$ at some point, and θ^{NC} stays finite: $\tau \leq 3/|\tilde{\theta}_0 - \theta_0^{\mathrm{NC}}|$; or
- the NCEC holds and $\tilde{\theta}_0 < 0$ at some point: $\tau \leq 3/|\tilde{\theta}_0|$.

Lemma (null). Same two conditions with $R_{\mu\nu}k^\mu k^\nu$, and the bounds $\lambda \leq 2/|\tilde{\theta}_0 - \theta_0^{\mathrm{NC}}|$ and $\lambda \leq 2/|\tilde{\theta}_0|$ respectively.

- Conjugate points: with $\dot{A} = \tilde{B}A$, $\tilde{\theta} = \dfrac{\mathrm{d}}{\mathrm{d}\tau}\ln|\det A|$, so q is conjugate to p iff $\tilde{\theta} \to -\infty$ at q.

Extremal Length under Non-Commutativity

- Geodesic deviation is modified:

$$U^\nu \nabla_\nu \big(U^\lambda \nabla_\lambda V^\mu\big) = -R^\mu{}_{\nu\lambda\rho} V^\nu U^\lambda U^\rho + U^\nu \nabla_\nu \big(\theta^{\mathrm{NC}} V^\mu\big).$$

- Choosing the parametrization $f = 1$ along the geodesic and X^μ orthogonal to T^μ, the first variation of proper time is unchanged, while the second variation of proper time becomes

$$\begin{aligned}\left.\frac{d^2\tau}{d\alpha^2}\right|_{\alpha=0} &= \int_a^b X^\nu \Big\{ T^\rho \nabla_\rho (T^\mu \nabla_\mu X_\nu) - T^\rho \nabla_\rho (\theta^{\mathrm{NC}} X_\nu) + R^\sigma{}_{\rho\mu\nu} T^\mu T_\sigma X^\rho \Big\}\, dt \\ &= \int_a^b X^\nu (\mathcal{O}X)_\nu \, dt.\end{aligned}$$

Theorem 3.8

Let γ be a smooth timelike curve connecting two points $p, q \in M$. Then the necessary and sufficient condition that γ locally maximizes the proper time between P and q over smooth one-parameter variations is that γ be a geodesic with no point conjugate to p between p and q.

Curvature and Conjugate Points under Non-Commutativity

- The extrinsic curvature of a hypersurface Σ is now

$$K = \theta = \tilde{\theta} - \theta^{\mathrm{NC}},$$

so conjugacy to Σ under the NCEC requires $K < -\theta^{\mathrm{NC}}$.

Lemma 3.9

Let U^μ be the tangent field of a hypersurface-orthogonal timelike geodesic congruence. Let Σ be a spacelike hypersurface with extrinsic curvature K. Then there exists a point p conjugate to Σ along the geodesic γ orthogonal to Σ and passing through q, if either of the following conditions is satisfied:

- $R_{\mu\nu}U^\mu U^\nu \geq 0$, $K < 0$ and θ^{NC} remains finite along the geodesic up to the focusing point. In this case, the point p conjugate to Σ exists within $\tau \leq 3/|K|$
- The NCEC, $R_{\mu\nu}U^\mu U^\nu \geq \mathcal{E}^{\mathrm{NC}}$, holds, $K < -\theta^{\mathrm{NC}}$. In this case, the point p conjugate to Σ exists within $\tau \leq 3/|K + \theta^{\mathrm{NC}}|$

Non-Commutative Singularity Theorem I: Timelike Incompleteness

Theorem 3.11

Let $(M, g_{\mu\nu})$ be a globally hyperbolic spacetime. Suppose there exists a smooth (or, at least C^2) spacelike Cauchy surface Σ. Then, all past-directed timelike curves are incomplete if

- The SEC, $R_{\mu\nu}U^\mu U^\nu \geq 0$ holds and the extrinsic curvature of the Cauchy surface Σ satisfies $K \leq C < 0$ everywhere, and θ^{NC} remains finite along the geodesic up to the focusing point. In this case, no past timelike directed curve from Σ can have length greater than $3/|C|$
- The NCEC

$$R_{\mu\nu}U^\mu U^\nu \geq \frac{2\theta^{\mathrm{NC}}\tilde{\theta}}{3} + \frac{d\theta^{\mathrm{NC}}}{d\tau} - \frac{(\tilde{\theta})^2}{3}$$

holds, and the extrinsic curvature of the Cauchy surface Σ satisfies $K \leq C < -\theta^{\mathrm{NC}}$ everywhere. In this case, no past timelike directed curve from Σ can have length greater than $3/|C + \theta^{\mathrm{NC}}|$

Non-Commutative Singularity Theorem II: Null Incompleteness

Theorem 3.12

Let $(M, g_{\mu\nu})$ be a connected, globally hyperbolic spacetime with a noncompact Cauchy surface Σ. Suppose further that M contains a trapped surface T. Then, at least one inextensible future-directed orthogonal null geodesic from T is incomplete if,

- The Strong Energy Condition, $R_{\mu\nu}k^\mu k^\nu \geq 0$ holds, the maximum value of $\tilde{\theta} - \theta^{\mathrm{NC}}$ satisfies $\tilde{\theta}_0 - \theta^{\mathrm{NC}} < 0$, and θ^{NC} remains finite along the geodesic up to the focusing point. In this case, at least one inextensible future-directed orthogonal null geodesic from T has affine length no greater than $3/|\tilde{\theta}_0 - \theta_0^{\mathrm{NC}}|$
- The NCEC

  $$R_{\mu\nu}U^\mu U^\nu \geq \mathcal{E}^{\mathrm{NC}}$$

  holds, the maximum value of $\tilde{\theta}$ satisfies $\tilde{\theta}_0 < 0$. In this case, no past timelike directed curve from Σ can have length greater than $3/|\tilde{\theta}_0|$

Dark energy: a brief history

- **1988:** Ratra & Peebles developed quintessence, a slow rolling scalar field proposed as an alternative to a bare cosmological constant.
- **1998:** Riess et al showed that Type Ia supernovae reveal the accelerated expansion directly.
- **1999:** Caldwell introduced Phantom Dark Energy, which has an equation of state parameter $\omega = p/\rho < -1$.
- **2024-25:** DESI shows a mild but persistent preference for an evolving equation of state over Λ, the observational motivation for a geometric mechanism that predicts dynamics for free.

Specializing to FLRW

In a homogeneous and isotropic FLRW background the expansion is tied to the Hubble parameter,

$$\tilde{\theta} = 3H.$$

The modified Raychaudhuri equation becomes

$$3\dot{H} = -3H^2 + 2H\theta^{\mathrm{NC}} - \frac{(\theta^{\mathrm{NC}})^2}{3} - R_{\mu\nu}U^\mu U^\nu + \dot{\theta}^{\mathrm{NC}}.$$

Equivalently, we have

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \underbrace{\frac{2}{3}H\theta^{\mathrm{NC}} - \frac{(\theta^{\mathrm{NC}})^2}{9} + \frac{\dot{\theta}^{\mathrm{NC}}}{3}}_{\text{geometric source}}.$$

- No extra matter field, no modification of Einstein's equations.

Effective pressure

Absorbing the geometric terms into an effective pressure (with ρ and the Friedmann equation untouched),

$$P_{\text{eff}} = P - \frac{H\theta^{\text{NC}}}{6\pi G} + \frac{(\theta^{\text{NC}})^2}{36\pi G} - \frac{\dot{\theta}^{\text{NC}}}{12\pi G}.$$

Modified continuity equation

Combining the Friedmann and Raychaudhuri equation,

$$\dot{\rho} = -3H(\rho + P) + \frac{4\rho}{3}\theta^{\mathrm{NC}} - \frac{H(\theta^{\mathrm{NC}})^2}{12\pi G} + \frac{H\dot{\theta}^{\mathrm{NC}}}{4\pi G}.$$

The corrections separate into two classes:

Linearly Proportional to ρ

- The term $\frac{4}{3}\rho\,\theta^{\mathrm{NC}}$.
- A universal modification of the dilution of every fluid.

Non-Linearly Proportional to ρ

- Terms in $(\theta^{\mathrm{NC}})^2$ and $\dot{\theta}^{\mathrm{NC}}$.
- Cannot be directly designated to any conventional fluid: absorbed into the dark-energy sector.

Modified dilution of matter, radiation and curvature

For $i = r, m, k$:

$$\dot{\rho}_i + 3H(1+\omega_i)\rho_i = \frac{4\theta^{\mathrm{NC}}}{3}\rho_i \;\Rightarrow\; \rho_i(a) = \rho_{i,0}\, a^{-3(1+\omega_i)} \exp\Big[\int_{t_0}^{t} \tfrac{4}{3}\theta^{\mathrm{NC}}\,\mathrm{d}t'\Big].$$

Define

$$F = \exp\Big[-\int \tfrac{4}{3}\theta^{\mathrm{NC}}\,\mathrm{d}t\Big] \;\Rightarrow\; \rho_r \propto \frac{a^{-4}}{F}, \quad \rho_m \propto \frac{a^{-3}}{F}, \quad \rho_k \propto \frac{a^{-2}}{F}.$$

- Standard scaling laws are modified by a single universal factor $1/F$.

Early times: the Hubble tension

- During radiation domination, $H^2 \simeq \dfrac{8\pi G}{3}\rho_r$, so larger θ^{NC} at early times enhances ρ_r and the early-time H.
- The CMB tightly fixes the angular acoustic scale

$$\theta_s = \frac{r_s(a_*)}{\chi(a_*)} = \frac{\int_0^{a_*} \frac{c_s\,\mathrm{d}a}{a^2 H(a)}}{\int_{a_*}^{a_0} \frac{\mathrm{d}a}{a^2 H(a)}}.$$

- Faster early expansion $\Rightarrow$ smaller sound horizon r_s. To keep θ_s fixed, $\chi(a_*)$ must shrink, which requires a larger inferred H_0.
- $F < 1$ before recombination therefore shifts the CMB-inferred H_0 toward the late-time measured value, offering a geometric route to easing the Hubble tension.

The effective dark-energy sector

The evolution of $\dot{\rho}_{\rm DE}$ is given by,

$$\dot{\rho}_{\rm DE} - \frac{4\theta^{\rm NC}}{3}\rho_{\rm DE} + \frac{H(\theta^{\rm NC})^2}{12\pi G} - \frac{H\dot{\theta}^{\rm NC}}{4\pi G} = 0,$$

a first-order linear equation with formal solution

$$\rho_{\rm DE} = \frac{1}{F}\Big[\rho_{\rm DE}(t_0) - \frac{1}{12\pi G}\int_{t_0}^{t} F\left((\theta^{\rm NC})^2 - 3\dot{\theta}^{\rm NC}\right) H\,{\rm d}t\Big].$$

- The first factor is the universal dilution modification.
- The integral term continuously sources or depletes dark energy according to the evolution of the non-commutative geometry.
- Unlike matter, radiation and curvature, dark energy is not just a rescaled standard law: it acquires genuinely new dynamics.

Equation of state

Comparing with $\rho_{\rm DE} \propto a^{-3(1+\omega_{\rm DE})}$ gives an effective $\omega_{\rm DE}(t)$

$$\omega_{\rm DE} = -1 + \frac{\left(\frac{\dot{F}}{3HF} + \frac{1}{36\pi G\rho(t_0)}\left(\frac{45}{16}\dot{F}^2 - \frac{9}{4}\ddot{F}F\right)\right)}{1 - \frac{F}{12\pi G}\int\left(\frac{45\dot{F}^2}{16F} - \frac{9\ddot{F}}{4}\right)Hdt}.$$

- Dynamical dark energy emerges naturally as a purely geometric consequence of the non-commutative structure of spacetime, without requiring additional exotic matter scalar field or modifications to Einstein's equation.
- In line with recent observational indications from DESI.
- Canonical, phantom-like or Λ-like behavior all arise from a single geometric input, and so does the fate of the Universe.

To conclude

- Relaxed $[U, V] = 0$ for geodesic congruences on a smooth manifold, motivated by the Frobenius theorem: $[U, V]^\mu = \theta^{\mathrm{NC}} V^\mu$.
- The physical expansion shifts to $\tilde{\theta} = \theta + \theta^{\mathrm{NC}}$, giving a modified Raychaudhuri equation.
- SEC + finite θ^{NC} still gives focusing. Moreover, in certain regimes, the **NCEC** can hold even when the SEC is violated, yielding non-commutative singularity theorems.
- In FLRW, θ^{NC} splits the continuity equation into a universal dilution correction and a dynamical dark-energy sector, with no extra fields and unmodified Einstein equations.
- Positive θ^{NC} at early times shrinks the sound horizon and can ease Hubble tension.

Thank you.

Questions and discussion welcome.

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